

Biology All In One (BAIO) Learning System

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Chapter 9: Biotechnology – Principles and Processes

Worksheet · Polymerase Chain Reaction (PCR)

I. Multiple Choice Questions

- PCR stands for:
(A) Polymerase Chain Reaction
(B) Protein Coding Region
(C) Plasmid Cloning Reaction
(D) Primer Chain Replication
- What are the three steps of PCR, in order?
(A) Denaturation, Annealing, Extension
(B) Annealing, Extension, Denaturation
(C) Extension, Denaturation, Annealing
(D) Ligation, Denaturation, Extension
- Which PCR step separates double-helical DNA into single strands using high temperature?
(A) Annealing
(B) Extension
(C) Denaturation
(D) Elution
- Which PCR step allows primers to bind to the single-stranded DNA template?
(A) Denaturation
(B) Annealing
(C) Extension
(D) Ligation
- Which PCR step involves DNA polymerase synthesising a new DNA strand?
(A) Denaturation
(B) Annealing
(C) Extension
(D) Elution

II. Fill in the Blanks

- PCR stands for Polymerase Chain _____.
- The three steps of PCR, in order, are denaturation, _____, and extension.
- Taq polymerase is isolated from the thermophilic bacterium _____.
- During annealing, the reaction temperature is lowered to about _____ °C.
- DNA polymerase extends the primer in the _____ direction.
- PCR can generate up to about _____ identical copies of the target DNA.

III. Match the Following

Column A	Column B
1. Denaturation	a. Thermostable enzyme from <i>Thermus aquaticus</i>
2. Annealing	b. Primers bind to template DNA
3. Extension	c. Strand separation at high temperature
4. Taq polymerase	d. Synthesis of new DNA strand

IV. True / False

- PCR is carried out in vitro, outside a living cell. [True / False]

2. Taq polymerase is destroyed at the high temperatures used for denaturation. [True / False]
3. Annealing occurs at a higher temperature than denaturation. [True / False]
4. Primers are chemically synthesised oligonucleotides complementary to the template DNA. [True / False]
5. Each PCR cycle roughly doubles the number of DNA molecules present. [True / False]

V. 2 Marks Questions

1. Explain why Taq polymerase is preferred over ordinary DNA polymerase in PCR.

2. Describe what happens during each of the three steps of a PCR cycle.

VI. 3 Marks Questions

1. Why does PCR require two sets of primers rather than one?

2. Explain how PCR can generate up to a billion copies of DNA from a single template.

VII. 5 Marks Questions

1. Explain the three steps of one PCR cycle in detail, and describe the role of Taq polymerase in enabling repeated amplification of DNA.
